

Prajit Kumar Das
Curriculum Vitae, Aug 10, 2026

 prajitdas@gmail.com |  <https://prajitdas.github.io>

WORK EXPERIENCE

11/1/2021 – Present	Cisco Systems, Inc. 8135 Maple Lawn Blvd. Fulton, MD, 20759 Leader, Software Engineering	As a Security researcher and Software Engineering Leader in Cisco's Security Business Group, I lead a high-performing team of 20+ Software Developers, Technical Leads, and Product Owners, having personally hired and built two-thirds of the team. I focus on building secure, scalable, and innovative solutions for the Secure Firewall Management Center and beyond. I drive the design and delivery of next-generation AI-powered security solutions by strategically integrating Generative AI and Large Language Models (LLMs) for intelligent firewalls and advanced management tools. Leveraging my ongoing research in "PRvL: Quantifying the Capabilities and Risks of Large Language Models for PII Redaction," I ensure robust protection of sensitive data, maintain compliance, and strengthen customer trust in Cisco products and networks. My key responsibilities and achievements include: • Security & Core Platform: I implement SAML SSO and OAuth2-based authentication workflows, lead enhancements in serviceability and observability for SaaS environments, and deliver containerized security solutions for MSSP customers and small-to-medium businesses. • AI & Automation: I deploy and integrate AI agents using LLMs to streamline network security operations, build secure data pipelines for RAG models, and lead automation initiatives for quality tracking and continuous improvement. • Compliance & Risk Management: I lead FedRamp certification, Cisco Product Security Baseline compliance, and internationalization (g11n) and accessibility efforts.
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- Vulnerability Response: I led the remediation of Blast-RADIUS (CVE-2024-3596; NVD Critical, CVSS 9.0) for Cisco Secure Firewall Management Center, tracked publicly as Cisco defect CSCwk71817 and shipped across multiple Secure Firewall release trains, partnering with Cisco PSIRT and Talos on rapid mitigation that supported millions in contract approvals from partners.
- Infrastructure & Operations: I oversee device telemetry, Jenkins CI management, cloud onboarding for on-prem solutions, and data queue optimization.
- Auditing & Reporting: I drive improvements in product licensing usage auditing and user activity tracking.
- Leadership & Collaboration: I provide technical vision, strategic direction, and mentorship across multiple engineering teams, promoting innovation, compliance, and operational excellence.

Through my leadership, I shape secure, user-centric, and future-ready systems across Cisco's security portfolio.

04/27/2020 –
10/31/2021
Cisco Systems, Inc.
8135 Maple Lawn Blvd.
Fulton, MD, 20759
Senior Software Engineer,
Product Owner

Led teams that work on Identity (MFA, SAML SSO), Access (RBAC) Management, Open Authentication (OAuth) based integration solutions for Cisco's Firepower Management Center Product series and containerization of software solutions for Cisco's products.

Worked on product design, product delivery management, team execution of software development process, software evaluation to ensure high quality of delivered software.

01/20/2019 –
04/26/2020
Cisco Systems, Inc.
8135 Maple Lawn Blvd.
Fulton, MD, 20759
Software Engineer, Product
Owner

Led teams that work on Identity (MFA, SAML SSO) and Access (RBAC) Management solutions for Cisco's Firepower Management Center Product series.

Worked on product design specification, product delivery management, team execution of software development process.

09/18/2017 –
01/19/2018
Cisco Systems, Inc.
8135 Maple Lawn Blvd.
Fulton, MD, 20759
Software Engineer

Worked on design, development and testing of product code for Cisco.

08/29/2012 –
08/22/2017
Ebiquity Research Group
University of Maryland,
Baltimore County,

Worked on research projects in the Mobile Access Control domains by implementing custom Android OS.

	1000 Hilltop Cir., Baltimore, MD, 21250 Research Assistant	
06/02/2014 – 08/22/2014	Samsung Research America 75 W. Plumeria Dr., San Jose, CA, 95134 PhD Intern, Advanced Applied Research Lab	Worked on mobile privacy projects implemented an Android app to execute custom Access Control solutions for Samsung’s research division.
03/06/2013 – 08/30/2013	Apple Inc. 1 Infinitie Loop, Cupertino, CA, 95014 PhD Summer intern in the Siri team	Worked on Data analysis of Siri’s customer queries. Discovered categories of erroneous responses for Siri.
06/28/2012 – 08/28/2012 & 05/31/2011 – 08/30/2011	Symantec Corp. 6750 Alexander Bell Dr., Columbia, MD, 21046 Intern (two internships; same project)	Worked on automated test framework to ensure new code does not break previous functionality.
2007 – 2010	Tata Consultancy Services Ltd. Plot C, Block EP, Salt Lake Electronics Complex, Sec - V, Bidhan Nagar, Kolkata, West Bengal, India, Pin – 700091 Software Engineer	Worked on design and development for Oracle DB and big data projects. Worked as a team lead for team of five developers.

EDUCATION

Doctor of Philosophy	Computer Science	University of Maryland, Baltimore County, USA	Fall 2017
Bachelor of Engineering (with honors)	Information Technology	Jadavpur University, India	Fall 2007

Dissertation title: Context-Dependent Privacy and Security Management on Mobile Devices

Advisor: Anupam Joshi, Ph.D. **Co-advisor:** Tim Finin, Ph.D.

PUBLICATIONS

2025

Leon Garza, Anantaa Kotal, Aritran Piplai, Lavanya Elluri, **Prajit Kumar Das**, Aman Chadha. "PRvL: Quantifying the Capabilities and Risks of Large Language Models for PII Redaction" In Proceedings of the *7th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications*, Nov, 2025

2020

Sofia Dutta, Sai Sree Laya Chukkapalli, Madhura Sulgekar, Swathi Krithivasan, **Prajit Kumar Das**, and Anupam Joshi. "Context Sensitive Access Control in Smart Home Environments." In Proceedings of the *6th IEEE International Conference on Big Data Security on Cloud (BigDataSecurity 2020)*. 2020.

2017

Prajit Kumar Das. "Context-Dependent Privacy and Security Management on Mobile Devices." *PhD thesis, University of Maryland, Baltimore County*, 2017.

Prajit Kumar Das, Anupam Joshi, and Tim Finin. "Personalizing context-aware access control on mobile platforms". In Proceedings *2017 IEEE 3rd International Conference on Collaboration and Internet Computing (CIC)*, pages 107--116. IEEE, 2017.

Prajit Kumar Das, Anupam Joshi, and Tim Finin. "App behavioral analysis using system calls." In Proceedings *2017 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS): MobiSec 2017: Security, Privacy, and Digital Forensics of Mobile Systems and Networks*, Atlanta, USA, May 2017.

2016

Prajit Kumar Das, Anupam Joshi, and Tim Finin. "Capturing policies for fine-grained access control on mobile devices." In Proceedings *2nd IEEE International Conference on Collaboration and Internet Computing (CIC 2016)*, Pittsburgh, PA, USA, November 1-3, 2016, November 2016.

Prajit Kumar Das, Dibyajyoti Ghosh, Pramod Jagtap, Anupam Joshi, and Tim Finin. "Preserving User Privacy and Security in Context-Aware Mobile Platforms." In Book *Mobile Application Development, Usability, and Security*. IGI Global, 2017. 166-193.

Prajit Kumar Das, Abhay Kashyap, Gurpreet Singh, Cynthia Matuszek, Tim Finin, and Anupam Joshi. "Semantic knowledge and privacy in the physical web." In Proceedings of the *4th Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn 2016) co-located with the 15th International Semantic Web Conference (ISWC 2016)*, Kobe, Japan, October 17, 2016., volume ISWC 2016, 2016.

Sudip Mittal, **Prajit Kumar Das**, Varish Mulwad, Anupam Joshi, and Tim Finin. "CyberTwitter: Using twitter to generate alerts for cybersecurity threats and vulnerabilities." In Proceedings *International Symposium on Foundations of Open Source Intelligence and Security Informatics (FOSINT-SI 2016) co-located with the 2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2016)*, San Francisco, CA, USA, August 19-20, 2016, 2016.

Prajit Kumar Das, Sandeep Narayanan, Nitin Kumar Sharma, Anupam Joshi, Karuna P. Joshi, and Tim Finin. "Context-sensitive policy-based security in internet of things." In Proceedings *2016 IEEE International Conference on Smart Computing (SMARTCOMP 2016)*, St Louis, MO, USA, May 18-20, 2016, pages 1--6, 2016.

2015

Primal Pappachan, Roberto Yus, **Prajit Kumar Das**, Sharad Mehrotra, Tim Finin, and Anupam Joshi. "Mobipedia: Mobile applications linked data." In *Proceedings of the ISWC 2015 Posters & Demonstrations Track co-located with the 14th International Semantic Web Conference (ISWC-2015)*, Bethlehem, PA, USA, October 11, 2015., volume 1486, pages 2--5. CEUR Workshop Proceedings (CEUR-WS.org), 2015.

Primal Pappachan, Roberto Yus, **Prajit Kumar Das**, Sharad Mehrotra, Tim Finin, and Anupam Joshi. "Building a mobile applications knowledge base for the linked data cloud." In *Proceedings of the 1st International Workshop on Mobile Deployment of Semantic Technologies (MoDeST 2015) co-located with 14th International Semantic Web Conference (ISWC 2015)*, Bethlehem, Pennsylvania, USA, October 11th, 2015., volume 1486, pages 14--25. CEUR Workshop Proceedings (CEUR-WS.org), 2015.

2014

Primal Pappachan, Roberto Yus, **Prajit Kumar Das**, Tim Finin, Eduardo Mena, and Anupam Joshi. "A semantic context-aware privacy model for FaceBlock." In *Proceedings of the 2nd Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn 2014) co-located with the 13th International Semantic Web Conference (ISWC 2014)*, Trento, Italy, October 20, 2014., volume 1316, pages 64--72. CEUR Workshop Proceedings (CEUR-WS.org), 2014.

Roberto Yus, Primal Pappachan, **Prajit Kumar Das**, Tim Finin, Anupam Joshi, and Eduardo Mena. "Semantics for privacy and shared context." In *Proceedings of the 2nd Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn 2014) co-located with the 13th International Semantic Web Conference (ISWC 2014)*, Trento, Italy, October 20, 2014., volume 1316. CEUR Workshop Proceedings (CEUR-WS.org), 2014.

Roberto Yus, Primal Pappachan, **Prajit Kumar Das**, Eduardo Mena, Anupam Joshi, and Tim Finin. "Demo: FaceBlock: privacy-aware pictures for Google Glass." In *Proceedings of the 12th Annual International Conference on Mobile Systems, Applications, and Services, MobiSys'14*, Bretton Woods, NH, USA, June 16-19, 2014, page 366, 2014.

2013

Prajit Kumar Das, Anupam Joshi, and Tim Finin. "Energy efficient sensing for managing context and privacy on smartphones." In *Proceedings of the 1st Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn 2013) co-located with the 12th International Semantic Web Conference (ISWC 2013)*, Sydney, Australia, October 22, 2013., volume 1121. CEUR Workshop Proceedings (CEUR-WS.org), 2013.

Prajit Kumar Das, Dibyajyoti Ghosh, Anupam Joshi, and Tim Finin. "ACM HotMobile 2013 poster: An energy efficient semantic context model for managing privacy on smartphones." *Mobile Computing and Communications Review*, 17(3):34--35, 2013.

NOTABLE CITATIONS

Independent researchers worldwide have cited and built on my work (Google Scholar: 520 citations, h-index 8, as of August 2026). Selected examples:

Tyagi, Jinwala, and Bhattacharjee (2024). *International Journal of Ad Hoc and Ubiquitous Computing*. Reproduced and extended the context-ontology design (Time/Interval and Activity classes) of my PALS context-sensitive access control work in their decentralized IoT access-control framework.

Shakarami and Sandhu (2021). *ACM Workshop on Secure and Trustworthy Cyber-Physical Systems (SAT-CPS '21)*. Situated their role-based smart-home administration model within the line of context-sensitive smart-home access-control research established by PALS.

Rahman, Mahdavi-Hezaveh, and Williams (2023). *ACM Computing Surveys*. Survey of automated cyber-threat-intelligence extraction that treats CyberTwitter as an established semantic-web CTI method, describing its RDF intelligence representation and ontology-based alert generation.

Wang et al. (2025). *Scientific Data (Nature Portfolio)*. Used CyberTwitter as a benchmark predecessor dataset in their comparison of open-source cybersecurity corpora for connected-autonomous-vehicle threat intelligence.

He et al. (2024). *IEEE Transactions on Information Forensics and Security*. Cited my system-call feature-extraction method ("App behavioral analysis using system calls") as prior work underpinning their BR-HIDF host intrusion-detection framework.

Shu, Zheng, and Hui (2018). *ACM Multimedia Systems Conference (MMSys '18)*. Their Cardea system builds directly on FaceBlock's visual-privacy premise, extending it into context-aware photo capture and sharing.

SERVICE TO PROFESSION

2026

Member, Artifact Evaluation Committee, 24th IEEE International Conference on Pervasive Computing and Communications (PerCom 2026)

2025

Program Committee Member, Workshop on Exploring Security, Trust, Ethics, and Data-centric issues in Autonomous Robotics Systems (STEADY), IEEE BigData 2025

Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC) — Reviewed multiple manuscripts including "AdaptiveShield," "SOOM," and "DP-Trojan"

Reviewer, European Symposium on Research in Computer Security (ESORICS 2025)

Reviewer Social Network Analysis and Mining

Reviewer, Qeios (Open Peer Review Platform)

2017

Program Committee Member & Reviewer, Industrial Knowledge Graphs Workshop (IKG), co-located with the 9th International ACM Web Science Conference

Reviewer, IEEE Transactions on Services Computing (TSC) — *Review of revised manuscript*

2016

Reviewer, ACM CHI Conference on Human Factors in Computing Systems (CHI 2017)

Reviewer, IEEE Transactions on Services Computing (TSC)

Sub-reviewer, 25th International World Wide Web Conference (WWW 2016)

2014

Program Committee Member & Additional Reviewer, 4th International Conference of Emerging Applications of Information Technology (EAIT 2014)

2012

Sub-reviewer, 76th IEEE Vehicular Technology Conference (VTC 2012-Fall)

RESEARCH EXPERIENCE

- 2025 **PRVL** (Privacy-Preserving Generative AI): Developed a framework to empirically quantify privacy leakage when Large Language Models (LLMs) perform Personally Identifiable Information (PII) Redaction. Introduced the Sensitive Privacy Violation (SPriV) score to establish a foundation for secure AI adoption in compliance-heavy U.S. industries (e.g., HIPAA). Published in the 7th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS 2025).
- 2016 **Mithril** Access Control (PhD dissertation project): Developed a framework for capturing user access control policies that are fine-grained, context-sensitive and are represented using Semantic Web technologies and thereby managing access control decisions for user data on mobile devices. Funded by NSF grants 0910838, 1228198 and MURI award FA9550-08-1-0265 from the Air Force Office of Scientific Research.
- 2016 **Carlton** (Google Physical Web project): Developed a Google Physical Web based context-aware privacy-managed question-answering system. Funded by Google gift.
- 2015 **Heimdall** (second part of PhD dissertation): Developed a Machine Learning classifier to determine if a mobile application's behavior at the operating system level, was potentially malicious versus benign.
- 2015 **Mobipedia** (A Mobile Applications Knowledge Base): Built a Linked data cloud containing knowledge of mobile application data, and developer meta-data, permissions used (in android), similar apps etc. Funded by NSF grants 0910838, 1228198.
- 2014 **FaceBlock** (Privacy-Aware Pictures): Built a proof-of-concept system capable of taking Privacy-Aware Pictures while using Google Glass. Funded by NSF grants 0910838 and MURI award FA9550-08-1-0265 from the Air Force Office of Scientific Research.
- 2014 Samsung Research America, Mobile privacy research internship: Created a privacy control app that protected users' data privacy. Focused on system level content providers and redirected app data requests, like requests made to *ContactsProvider* and responded with obfuscated or faked data.
- 2013 Apple Inc., Data analytics research internship in the Siri team: Identified areas of accuracy improvement for Siri. Carried out data analysis on user queries and Siri responses.

TALKS

- 2017 Paper presented “App behavioral analysis using system calls.” At the 2017 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS): MobiSec 2017: Security, Privacy, and Digital Forensics of Mobile Systems and Networks, May 1, 2017
- 2016 Paper presented “Capturing policies for fine-grained access control on mobile devices.” At the 2nd IEEE International Conference on Collaboration and Internet Computing (CIC 2016), Nov 1 – 3, 2016
- 2016 Paper presented “Semantic knowledge and privacy in the physical web.” At the 4th Workshop on Society, Privacy and the Semantic Web - Policy and Technology (PrivOn 2016) co-located with the 15th International Semantic Web Conference (ISWC 2016), Oct 17 – 21
- 2016 Paper presented “Context-sensitive policy-based security in internet of things.” At the 2016 IEEE International Conference on Smart Computing (SMARTCOMP 2016), May 18 – 20
- 2016 Short 3-minute talk “Energy Efficient, Context-Dependent Privacy and Security Management on Mobile Devices” At the 38th Annual UMBC Graduate Research Conference 2016, Mar 23
- 2015 Poster presented “Mobipedia: Mobile applications linked data.” At the 14th International Semantic Web Conference (ISWC 2015), Oct 11 – 15
- 2015 Research presented “Is your personal data at risk? App analytics to the rescue.” At the UMBC Cyber Defense Lab Meeting, Oct 23
- 2013 Poster presented “An energy efficient semantic context model for managing privacy on smartphones.” At the 14th International workshop on Mobile Computing Systems and Applications (HotMobile 2013), Feb 26 – 27

TEACHING EXPERIENCE

Guest Lecturer for Wireless Mobile Systems Security (Undergrad)	Spring 2016
Part-time TA for Wireless Mobile Systems Security (Undergrad)	Spring 2016
Guest Lecturer for Semantic Web (Grad)	Spring 2016
TA for Software Engineering (Grad)	Spring 2012
TA for Data Structures (Undergrad)	Fall 2010 - Spring 2012

EXTRACURRICULAR UNIVERSITY SERVICE

- 2014 UMBC ACM Student Chapter Chair (2013-2014) responsible for organizing Tech. Talks by visitors to the university and other department social activities like “Hi-Tea Competition.”
- 2013 UMBC ACM Student Chapter Treasurer (2012-2013) maintained budget for the chapter.
- 2013 Co-organizer / graduate student coordinator of hackUMBC (2013), the first hackathon at UMBC.

MEDIA COVERAGE

FaceBlock was mentioned in Nature Article: “What could derail the wearables revolution?” by Kat Austen as a potential solution created for handling privacy issues in the data-drenched world of Wearable Computing.

AWARDS

Awarded the NSF Student Travel Award of 2270 USD for presenting paper at the PrivOn workshop in the 15th International Semantic Web Conference (ISWC), 2016.

Awarded the NSF Student Travel Award for presenting paper at the SmartSys workshop in the 2nd IEEE International Conference on Smart Computing (SMARTCOMP), 2016.

Awarded the NSF Student Travel Award of 1427 USD for presenting paper at the MoDeST workshop in the 14th International Semantic Web Conference (ISWC), 2015.

Awarded the NSF Student Travel Award of 2200 USD for presenting paper at the PrivOn workshop in the 13th International Semantic Web Conference (ISWC), 2014.

TECHNICAL SKILLS

Platforms	Android, Linux, Python-Selenium based automation platform
Languages	Java, Python, Go, Bash
Tools and Stacks	IntelliJ Suite, LAMP, Git, Scikit-learn, Beautiful Soup, Plotly, Weka, Perforce, LaTeX
Back-end	MySQL, Oracle, NoSQL, MongoDB
Front-end	PHP, Bootstrap, HTML5, JavaScript, CSS, React
Domain knowledge	Identity & Access Management, Multi-Factor Authentication, Security Assertion Markup Language – Single Sign-On, Role-Based Access Control, Open Authentication, Cybersecurity, Semantic Web, Machine Learning, Mobile Computing, Mobile Security & Privacy, Enterprise Networking Software, Containerization of Software Solutions

TEACHING AREAS

Specialization	Mobile Computing, Computer Security and Privacy, Identity & Access Management
Core graduate	Operating Systems, Computer Networks, Database Management Systems
Core Undergrad	Java, Data Structures, Discrete Structures, Principles of Programming Languages

LANGUAGES

English	Fluent in all forms of communication
Bengali	Native language, fluent in all forms of communication
Hindi	Fluent in all forms of communication

****References available upon request****